



Review

Pesticide impacts on insect pollinators: Current knowledge and future research challenges

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Highlights

- Long-term global analysis of inter-regional trends of pesticide use presented
- Updated analysis of pesticides' impact on pollinators from different global regions
- Research challenges for the future pesticide policy towards pollinator conservation outlined

Abstract

With the need to intensify agriculture to meet growing food demand, there has been significant rise in pesticide use to protect crops, but at different rates in different world regions. In 2016, the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES) global assessment on pollinators, pollination and food production identified pesticides as one of the major drivers of pollinator decline. This assessment highlighted that studies on the effects of pesticides on pollinating insects have been limited to only a few species, primarily from developed countries. Given the worldwide variation in the scale of intensive agricultural practices, pesticide application intensities are likely to vary regionally and consequently the associated risks for insect pollinators. We provide the first long-term, global analysis of inter-regional trends in the use of different classes of pesticide between 1995 and 2020 (FAOSTAT) and a review of literature since the IPBES pollination assessment (2016). All three pesticide classes use rates varied greatly with some countries seeing increased use by 3000 to 4000% between 1995 and 2020, while for most countries, growth roughly doubled. We present forecast models to predict regional trends of different pesticides up to 2030. Use of all three pesticide classes is to increase in Africa and South America. Herbicide

use is to increase in North America and Central Asia. Fungicide use is to increase across all Asian regions. In each of the respective regions, we also examined the number of studies since 2016 in relation to pesticide use trends over the past twenty-five years. Additionally, we present a comprehensive update on the status of knowledge on pesticide impacts on different pollinating insects from literature published during 2016–2022. Finally, we outline several research challenges and knowledge gaps with respect to pesticides and highlight some regional and international conservation efforts and initiatives that address pesticide reduction and/or elimination.

Graphical abstract



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Introduction

The area of land under agriculture has expanded over recent decades in response to growing food demands (Ramankutty et al., 2018). Continuation of this trend is projected to lead to an additional land demand of 1510 Mha by 2030 (Lambin et al., 2013; Lambin and Meyfroidt, 2011). In parallel, the per capita availability of agricultural land has been shrinking (FAO, 2020²), mostly in the developing world. This demand for additional land and pressure to maintain agricultural output have driven the need to intensify agricultural production, with some regions undergoing more intensive agriculture relative to others (Zabel et al., 2019). Various environmental and economic shocks to food systems due to recent and ongoing geo-political turmoil (e.g., the Russian-Ukrainian war impacts on food, fuel, and fertilizer) and effects of climate change (e.g., droughts worldwide) have further added to this pressure. With increasing intensification, there has been concomitant growth in the use of agrochemicals particularly pesticides³ (i.e., insecticides, herbicides, and fungicides), although there is considerable variation in pesticide use across regions (Pellegrini and Fernández, 2018).

Worldwide, pesticides (along with other facets of land-use change and land-management intensification) are well established as being among the principal anthropogenic pressures linked to loss of insect populations (Osterman et al., 2019; Sánchez-Bayo and Wyckhuys, 2019; Straw et al., 2021, Straw et al., 2022; Straw and Brown, 2021a, Straw and Brown, 2021b; Dicks et al., 2021; Linguadoca et al., 2022; Tamburini et al., 2021a; although not all studies found negative effects, e.g., Tamburini et al., 2021b; Schwarz et al., 2022). Pesticide impacts on pollinating insects in particular have been reported consistently over recent decades both through empirical studies, meta-data analyses and syntheses (Cecala and Wilson Rankin, 2021; Douglas et al., 2022; Gill and Raine, 2014; IPBES, 2016; Kenna et al., 2019; Main et al., 2020; Raine and Rundlöf, 2024; Rundlöf et al., 2015; Stanley et al., 2016). Despite this scientific focus, various gaps in our understanding of pesticides' lethal and sub-lethal impacts on insect pollinators remain and were identified by the IPBES

assessment on Pollinators, Pollination and Food Production (IPBES, 2016, *ibid.* p.64). Considering the crucial role of pollinating insects in plant reproduction and the fact that approximately 85% of the main crop types around the world depend on insect pollination to varying degrees (Klein et al., 2007; Aizen et al., 2019), bridging these knowledge gaps is crucial to inform a more comprehensive pesticide risk assessment framework for safeguarding pollinating insects across the globe.

One major knowledge gap is the extent of regional variation of pesticide use and availability of standardized and/or accurate data for pesticides in different parts of the world (Dicks et al., 2021). To date research on pesticides and their impacts has been predominantly done in Europe and North America, with relatively little information from other world regions where conventional intensification of agriculture has gained momentum over the past half a century (Kuyper and Struik, 2014; Pretty et al., 2006). Given the inter-regional variation in levels of agricultural intensification and agrochemical use, the intensity and magnitude of pesticide application and associated risks are also likely to vary between regions. A recent expert-elicitation analysis in which experts scored the severity and magnitude of pressures on pollinators (Dicks et al., 2021) concluded that pesticides were among the most important causes of pollinator decline globally, but with variation in that pressure across regions. This risk to pollinators is likely growing in many world regions. This is because the latest generation of pesticides have been reported to have a greater Total Applied Toxicity (TAT).⁴ A recent meta-analysis of 381 pesticides used over the last twenty-five years that considered 1591 substances with specific acute toxicity threshold values for eight non-target species groups showed a marked increase in total applied toxicity (TAT) of insecticides since 2005 (Schulz et al., 2021). This study concluded that these increased toxicities are being driven by use of certain insecticides, such as highly toxic pyrethroids and neonicotinoids (Schulz et al., 2021).

In addition to the aforementioned regional variation, there is also a major bias in the selection of non-target species for pesticide-impact studies. The greatest focus has been on the western honey bee (*Apis mellifera*) (Dirilgen et al., 2023) as a model species, and to a lesser extent, on a few species of, often managed, bumble bees (*Bombus* spp. e.g., Tamburini et al., 2021a; Wintermantel et al., 2022) and stingless bees (Bernardes et al., 2018b; da Costa Domingues et al., 2020; De Oliveira Ferreira et al., 2020; Farder-Gomes et al., 2021a, Farder-Gomes et al., 2021b). Relatively little is known about pesticide impacts on most wild, unmanaged bee species and of those that exist, these mainly focus on the Megachilid bee *Osmia* spp. (Al Nagggar and Paxton, 2021; Li and Rangel, 2018; Mallinger et al., 2015; Milone et al., 2021; Park et al., 2015; Rundlöf et al., 2015; Schwarz et al., 2022; Williams et al., 2015; Woodcock et al., 2017; Zhu et al., 2014) and *Megachile rotundata* (Ansell et al., 2021; Piccolomini et al., 2018; Pitts-Singer and Barbour, 2017). In addition, still fewer have studied pesticide impacts on non-bee insect pollinators (Pisa et al., 2015; Serrão et al., 2022; Uhl and Brühl, 2019).

Relatively little research has been dedicated to assessing impacts of different classes of pesticides (but see Tosi and Nieh, 2019; Schwarz et al., 2022; Tamburini et al., 2021a; Tamburini et al., 2021b; Tosi et al., 2022). Most attention involving pesticide impacts on pollinators have tended to focus on insecticides, with relatively less known about how herbicides, fungicides, and especially acaricides and their co-formulants, impact pollinator health (Straw et al., 2022). Research conducted on the different insecticide classes has also been skewed, with most recently a focus on neonicotinoids (a family of systemic insecticides), largely stimulated by prominent societal and political debate (Godfray et al., 2015; Pisa et al., 2015). In addition, published studies have tended to report mostly on lethal toxicity endpoints, with less information available on sublethal impacts of pesticide exposure (Gill and Raine, 2014; Siviter et al., 2018a; Siviter et al., 2018b; Tosi et al., 2022; Tosi and Nieh, 2019). It is also extremely important to understand the synergistic impacts of different pesticide classes, including insecticides, herbicides, and fungicides, as well as their co-formulants and 'inert' ingredients (Mullin, 2015; Straw et al., 2022), and how these interact with other stressors, such as pathogens (González-Varo et al., 2013; Grassl et al., 2018; O'Neal et al., 2018; Schwarz et al., 2022; Siviter et

al., 2021; Tamburini et al., 2021a, Tamburini et al., 2021b; Tosi and Nieh, 2019; Tosi et al., 2022). Investigating these interactions is crucial for pollinator survival, and we are only now beginning to understand some of these complex dynamics. Even more scant, are studies examining the effect of different pesticide classes and their interaction with other factors, such as availability of flower/nutritional food resources (see Tosi et al., 2017; Klaus et al., 2021; Wintermantel et al., 2022). There is also little information on the long-term and cumulative effects impacting on different life-stages and castes (Tosi and Nieh, 2019).

In this paper, we first compare trends in the use of pesticides (insecticides, herbicides, fungicides) across different global regions from the 1995 to 2020 using the latest available global data from FAOSTAT⁵ (countries with non-availability or inconsistency in data, either in one or both the years, have been excluded). We then use Automatic autoregressive integrated moving average (Auto ARIMA) models (Hyndman and Khandakar, 2008) to provide a forecast of likely trends in pesticide use (insecticides, herbicides, fungicides) in different global regions up to 2030.

We then highlight literature published since the 2016 IPBES pollination assessment that examines the effects of pesticides on diverse pollinator species. Literature since 2016 till 2022 was searched exhaustively in Scopus and Google Scholar using strings of relevant key words, and based on these analyses, we finish by discussing the future research trajectory and highlight a few global policy and conservation initiatives addressing pesticide use.

Section snippets

Trends of pesticide use across global regions

Pesticide use (insecticides, herbicides, and fungicides) has varied markedly across different countries, even within a global region, over the last two decades. The intensity of pesticide impact depends on application rate (kg/ha) and the toxic load of different pesticide formulations (Douglas et al., 2022). It is therefore prudent to consider both elements when judging the risk from pesticide use, particularly given the shift towards low application rate-high toxicity pesticides. However, for ...

Impacts of pesticides on pollinators

Pesticides can have both lethal and sublethal effects on pollinators, either directly or indirectly via effects on habitat, nesting and forage resources (IPBES, 2016). The following sections primarily cover information on the direct lethal and sublethal impacts on bees of insecticides, herbicides, and fungicides. Most of this reviewed evidence is from analyses published after the IPBES, 2016 assessment (albeit sometimes containing evidence from before that date but republished in data ...

Advancement of knowledge and existing gaps

Our understanding of the effects of pesticides on pollinators, particularly bees, was the focus of our review. New information is emerging from other global regions beyond North America and Europe, as well as on previously under-researched species of pollinators. However, the overall geographic skew in information towards the developed economies remains and fewer pesticide impact studies on pollinators are being done in major world regions where pesticide use has been undergoing dramatic ...

Towards a comprehensive pesticide risk assessment framework

In a first ever global analysis of pesticide pollution risk, with 92 active pesticide ingredients across 168 countries, Tang et al. (2021) found that 31% of countries with agricultural land are at high risk from pesticide pollution and of those countries, 34% are found in biodiversity rich areas. The most vulnerable areas include South Africa, China, India, Australia, and Argentina. A key outcome of our analysis shows pesticide use has been growing in these regions too. A number of regions ...

An outlook on pesticide policies

Various policies in the last decade have focused on reducing (i.e., involving appropriate use of pesticide application) or eliminating the use of pesticides. Existing efforts to monitoring and track pesticide use, such as the US Geological Survey's National Pesticide Use Maps (USGS) are in danger of being limited or reduced due to budgetary constraints (Gewin, 2023). Policies or legislation that reduce or fully discontinue the use of pesticides should provide users with safe and effective ...

Future research and policy needs

Although there has been a concerted effort to increase the basic information and accurate reporting of pesticides and impacts on pollinators in regions outside of North America and Europe – many large gaps still exist despite several major studies from some regions in Asia and South America since 2016. Research on insecticides other than neonicotinoids has also gained some momentum and several studies have looked into the impacts of a number of alternative “environmentally safer” pesticides ...

CRedit authorship contribution statement

P. Basu: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **H.T. Ngo:** Writing – review & editing, Writing – original draft, Conceptualization. **M.A. Aizen:** Writing – review & editing. **L.A. Garibaldi:** Writing – review & editing, Methodology. **B. Gemmill-Herren:** Writing – review & editing. **V. Imperatriz-Fonseca:** Writing – review & editing. **A.M. Klein:** Writing – review & editing. **S.G. Potts:** Writing – review & editing, Conceptualization. **C.L.** ...

Declaration of competing interest

There are no financial and personal relationships with other people or organizations that could inappropriately influence (bias) this work and the authors have nothing to declare. ...

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